Comprehensive Structural Engineering Rebuild Syllabus

27 July 2026

Purpose and intended use

For a temporary works engineer rebuilding permanent-works fundamentals, Eurocode fluency, and long-term capability for tightly scoped structural work.

This is a multi-year syllabus, not a promise that you need to finish everything before becoming useful. Its purpose is to give you a complete map, with a realistic sequence and clear proof that you understand each stage.

At 2–3 focused hours per week, expect: • Year 1: structural mechanics foundation + basic Eurocode workflow + a straightforward EC3 beam design. • Years 2–3: deeper steel, timber, masonry, residential alterations, basic geotechnics/foundations, and concrete fundamentals. • Years 3–4+: integrated projects, analysis/model judgement, more complex stability, connections, professional practice, and building a portfolio of reviewed work.

Your career experience is the other half of the programme. Study gives you the theory; your work in temporary works, site, construction staging, and design coordination gives you the practical context.

Study rules

For every topic, do not move on until you can do all five:

1. Explain it plainly

No equations at first. Explain what is physically happening.

1. Sketch it

Draw load paths, support conditions, deformed shape, compression/tension zones, and failure mode.

1. Calculate a simple case by hand

No software and no copied method.

1. Check it against a reliable source

A textbook worked answer, recognised professional guidance, or a checked/anonymised office example.

1. State assumptions and limits

What must be true for the calculation to be valid? What information is missing? What could change the answer?

For every significant example, prepare a short calculation note: • brief and sketch; • structural arrangement and load path; • design assumptions; • loads/actions; • analysis; • checks; • detailing/restraint requirements; • conclusion; • limitations and questions requiring review.

Your reference library

First-principles resources

1. Structural Analysis — R.C. Hibbeler

Use for: • equilibrium; • reactions; • trusses and frames; • shear-force and bending-moment diagrams; • deflection; • indeterminate analysis later.

1. Mechanics of Materials — R.C. Hibbeler

Use for: • stress and strain; • axial deformation; • torsion; • second moment of area; • bending stress; • beam deflection; • shear stress; • buckling.

These two should be your core problem books in Year 1.

1. Jeff Hanson videos

Use as a supporting explanation tool for: • statics; • free-body diagrams; • reactions; • SFDs/BMDs; • centroids; • second moment of area; • stress/strain; • beam bending; • deflection; • torsion; • Euler buckling.

Do not treat videos as completed learning. Use this sequence: Attempt problem → watch explanation only if needed → redo problem without help → explain physical behaviour in writing.

UK design and practical resources

Steel • SteelConstruction.info • SCI design guides, where your employer/library provides access • SCI Steelwork Design Guide to Eurocode 3 • SCI P363 Steel Building Design: Design Data, if available

Search terms: • site:steelconstruction.info lateral torsional buckling • site:steelconstruction.info Eurocode 3 beam design • site:steelconstruction.info bracing steelwork • site:steelconstruction.info section classification

Concrete • Reinforced Concrete Design to Eurocode 2 — Mosley, Bungey and Hulse • The Concrete Centre • Concrete Centre Eurocode 2 design guides

Search: • site:concretecentre.com EC2 beam design • site:concretecentre.com reinforced concrete flexure shear • Eurocode 2 RC beam worked example

Timber • TRADA • Wood Campus • Timber Engineering — Thelandersson and Larsen, later if you want deeper theory

Search: • site:woodcampus.co.uk Eurocode 5 beam • TRADA timber beam design Eurocode 5 • timber notch shear design guidance UK

Masonry • IStructE resources where available • UK masonry technical guidance and manufacturer resources • Eurocode 6 resources through your employer/university library

Search: • masonry beam bearing padstone design UK • masonry lateral restraint structural engineering • opening in load bearing wall calculation example UK

Soil and foundations • Craig's Soil Mechanics • Principles of Geotechnical Engineering — Braja M. Das • ICE Manual of Geotechnical Engineering, where available • CIRIA guidance, through employer access if possible

Search: • effective stress soil mechanics worked example • Eurocode 7 pad foundation worked example UK • "factual ground investigation report" pdf UK • site:gov.uk ground investigation report pdf

Standards

Use your employer's current controlled copies or a legitimate university/library/BSI route. Verify the applicable editions, National Annexes, and project requirements.

Core standards to become familiar with: • EN 1990: Basis of structural design • EN 1991: Actions on structures • EN 1992: Concrete • EN 1993: Steel • EN 1995: Timber • EN 1996: Masonry • EN 1997: Geotechnical design • BS 5975 and your temporary works procedures • National Annexes and any UK implementation requirements

Do not try to memorise standards. Learn how they are structured and how to navigate them.

Weekly structure

At 2–3 hours per week:

Weekly pattern

Session 1 — Concept and sketching: 45–60 minutes • Read one section of textbook/guidance or watch one focused video. • Make a one-page note in your own words. • Draw diagrams.

Session 2 — Hand problem: 45–60 minutes • Attempt one or two problems without the solution. • Show all units and assumptions.

Session 3 — Check and application: 45–60 minutes • Compare with the authoritative worked solution. • Correct your work. • Write three bullets: • what you misunderstood; • what physical behaviour the maths describes; • how it relates to a real temporary/permanent works situation.

Every fourth week Do not start new content. Instead: • redo an old problem from memory; • create one short calculation note; • revisit one live/historic work example; • note questions for a senior engineer.

Module 0 — Engineering maths, units and physical quantities

Commented [UI1]: We need to cover precast, backpropping lifting, gfeotechnical reports cat 2 and ask for straightforward exmaples and learn how to do it

Suggested time: 4–6 weeks

Why it matters Almost every serious calculation error is either: • a wrong model; • a wrong assumption; • or a units/conversion error.

You need units to feel automatic.

Learn

Units • N, kN, MN • mm, m • mm^2, m^2 • mm^4, m^4 • kN/m • kN/m^2 • N/mm^2 and MPa • GPa • kg/m^3 and kN/m^3 • radians and degrees

Maths • scientific notation; • algebra and rearranging equations; • ratios and proportions; • trigonometry; • Pythagoras; • simultaneous equations; • graph gradients and areas; • basic differentiation/integration meaning.

Physical quantities • mass versus weight; • force; • moment; • distributed load; • stress; • strain; • stiffness; • density; • pressure.

Worked examples

Example 0.1 — Unit conversion Convert: • 200 GPa into N/mm^2; • 25 kN/m^2 into N/mm^2; • 5 kNm into Nmm; • 7.5 kN/m into N/mm.

Example 0.2 — Resolving an inclined force A 20 kN tie is inclined at 35° to horizontal.

Calculate horizontal and vertical components.

Example 0.3 — Resultant of UDL A 6 m beam carries 8 kN/m.

Calculate the equivalent point load and where it acts.

Resources • Jeff Hanson: statics introduction and force vectors. • Hibbeler Structural Analysis: opening statics chapters. • EngineeringStatics.org: free statics exercises.

Gate to move on You can: • convert all common structural units; • resolve forces without prompts; • calculate moments correctly; • replace a UDL with its resultant; • check whether an equation is dimensionally sensible.

Module 1 — Statics, equilibrium and load paths

Suggested time: 8–10 weeks

Learn

Free-body diagrams • isolate the structure/member; • show all external loads; • show reactions; • choose axes; • label dimensions and load positions.

Equilibrium • ΣFx = 0 • ΣFy = 0 • ΣM = 0

Supports • roller support; • pin support; • fixed support; • internal hinge; • spring support, later; • what each permits and restrains.

Load paths • roof to walls/beams; • floors to beams/walls; • beams to columns/masonry; • columns/walls to foundations; • foundations to soil.

Tributary areas • floor loading; • roof loading; • line loads to supports; • load sharing assumptions.

Stability • stable and unstable systems; • determinate and indeterminate structures; • why a stable load path is not enough if lateral restraint is missing.

Worked examples

Example 1.1 — Beam reactions under UDL A 6 m simply supported beam carries 10 kN/m.

Calculate: • total load; • left reaction; • right reaction; • explain why symmetry applies.

Example 1.2 — Beam reactions under point load A 5 m beam carries 20 kN at 2 m from the left support.

Calculate reactions using moments about each support.

Example 1.3 — Tributary floor load A floor spans 4 m between two supporting beams. The beams are 5 m long. Floor actions are: • permanent action: 1.0 kN/m²; • imposed action: 1.5 kN/m².

Calculate characteristic line loads to each beam, excluding beam self-weight.

Example 1.4 — Real-work load path Take a falsework arrangement you understand. Draw: wet concrete → formwork → primary beams → secondary beams → heads → props/towers → baseplates → soleboards → ground.

Then list what could interrupt the load path.

Resources • Hibbeler Structural Analysis: reactions and equilibrium sections. • Jeff Hanson: statics, equilibrium, reactions. • Your own drawings, RAMS, temporary works calculations, or system manuals.

Gate You can: • draw a clean free-body diagram; • calculate reactions on standard determinate beams; • calculate tributary loads; • identify a complete vertical load path; • identify the lateral stability system; • explain why a padstone, bearing, connection, or foundation may govern rather than the beam.

Module 2 — Internal actions, shear force and bending moment

Suggested time: 8–10 weeks

Learn • axial force; • shear force; • bending moment; • sign conventions; • cutting a member to reveal internal actions; • SFDs; • BMDs; • load–shear–moment relationships; • sagging/hogging; • deflected shapes; • tension and compression zones.

Physical understanding • Axial tension pulls a member apart. • Axial compression pushes it together and can lead to buckling. • Shear tries to slide one part past another. • Bending curves a member: one side is in tension, the other in compression. • Maximum bending often occurs where shear passes through zero. • Maximum shear is often near supports, but the actual loading arrangement matters.

Worked examples

Example 2.1 — Simply supported UDL Repeat the 6 m, 10 kN/m beam.

Draw: • SFD; • BMD; • deformed shape; • tension/compression faces.

Confirm: • Vmax = wL/2 • Mmax = wL²/8

Example 2.2 — Cantilever under UDL A 2 m cantilever carries 5 kN/m.

Calculate: • fixed-end shear; • fixed-end moment; • SFD; • BMD; • which face is in tension at the fixed end.

Example 2.3 — Point load beam Use the 5 m beam with the 20 kN point load.

Draw SFD/BMD and find the maximum moment from the diagram.

Resources • Hibbeler Structural Analysis: shear and bending moment chapters. • Jeff Hanson: shear-force and bending-moment videos. • EngineeringStatics.org.

Gate You can: • draw SFD/BMDs without copying; • calculate maximum moment/shear; • predict the diagram before calculating it; • identify tension/compression zones; • explain why a beam may need shear checks close to supports even where bending moment is low.

Module 3 — Stress, strain, stiffness and material behaviour

Suggested time: 6–8 weeks

Learn

Stress and strain • normal stress: $\sigma = F/A$ • strain: $\varepsilon = \Delta L/L$ • stress–strain curves; • elastic range; • yield; • plastic deformation; • ultimate strength; • fracture.

Stiffness • axial deformation: $\Delta L = PL/AE$ • Young's modulus E; • shear modulus G; • Poisson's ratio ν; • relationship for isotropic materials: $E = 2G(1+\nu)$

Materials • structural steel; • reinforced concrete; • timber; • masonry; • ductile versus brittle behaviour; • creep, shrinkage, fatigue: introductory awareness.

Worked examples

Example 3.1 — Steel tie A steel tie: • length: 3,000 mm; • area: 1,000 mm^2; • tension force: 100 kN; • E = 200,000 N/mm^2.

Calculate: • stress; • strain; • extension.

Example 3.2 — Steel versus aluminium stiffness Two bars have the same area, length, and applied force. One is steel and one is aluminium.

Calculate/compare extensions using representative Young's moduli.

Example 3.3 — Material comparison note Write half a page explaining: • why concrete works well in compression; • why RC needs reinforcement; • why steel may fail by buckling rather than material crushing; • why timber properties vary with grain direction and moisture.

Resources • Hibbeler Mechanics of Materials: axial loading and stress/strain. • Jeff Hanson: Mechanics of Materials. • Concrete Centre introductory RC resources. • Wood Campus introductory timber material resources.

Gate You can: • calculate stress, strain, and extension; • distinguish strength from stiffness; • explain E and G; • explain yielding versus fracture; • explain why a steel member's strength does not guarantee its stability.

Module 4 — Section properties, bending theory and member shape

Suggested time: 10–12 weeks

This is one of your most important modules.

Learn

Geometry and centroids • area; • centroid; • neutral axis; • symmetrical and unsymmetrical shapes; • composite areas.

Second moment of area • I; • major/minor axes; • parallel-axis theorem; • why I depends on axis; • units of I: mm^4.

Section modulus • elastic section modulus: $Z = I/y$ • elastic bending stress: $\sigma = My/I = M/Z$ • plastic section modulus; • elastic versus plastic stress distributions; • plastic hinge concept.

Radius of gyration $i = \sqrt{(I/A)}$

Understand its relevance to buckling later.

Why I-sections work • bending stresses are greatest furthest from the neutral axis; • flanges resist most bending; • web keeps flanges apart and carries shear; • material near neutral axis contributes less to bending resistance; • depth generally improves bending stiffness/resistance strongly.

Major and minor axes For a typical universal beam: • major axis = strong bending axis; • minor axis = weak bending axis; • orientation and restraint matter.

Worked examples

Example 4.1 — Rectangle section properties For a 75 mm × 225 mm rectangular timber section: $I = bd^3/12$

Calculate: • I; • Z; • bending stress for M = 8 kNm.

Repeat with depth increased to 300 mm.

Write what changed: • bending stress; • stiffness; • likely deflection; • practical limitations.

Example 4.2 — Parallel-axis theorem Calculate the second moment of area of a simple built-up section made from two rectangular flanges and a web.

Example 4.3 — Major/minor axis of a UB Use a section table from SCI/SteelConstruction.info.

Choose one UB and identify: • major-axis I; • minor-axis I; • major/minor elastic section modulus; • ratio between them; • why using the section in the wrong orientation is dangerous.

Example 4.4 — Elastic versus plastic bending Sketch a rectangular steel section under increasing bending:

1. elastic stress distribution;
2. first yield at extreme fibres;
3. fully plastic stress block.

Then write why local buckling or LTB may prevent full plastic resistance being used.

Resources • Hibbeler Mechanics of Materials: centroids, moments of inertia, bending. • Jeff Hanson: centroid, moment of inertia, flexural stress. • SteelConstruction.info: steel section properties and section classification introductions.

Gate You can: • calculate I, Z, and bending stress for simple sections; • use parallel-axis theorem; • explain major/minor axes; • explain why I-sections are efficient; • distinguish I, Z, and plastic modulus; • explain why a deeper beam is usually more effective than simply using a higher steel grade when deflection governs.

Module 5 — Deflection, serviceability and vibration

Suggested time: 6–8 weeks

Learn • curvature; • flexural stiffness EI; • standard deflection formulae; • superposition; • span effects; • ULS versus SLS; • appearance, damage, usability, drainage, vibration; • creep/shrinkage in concrete; • moisture movement in timber; • floor vibration: introductory awareness.

Key understanding A beam can be safe against collapse but still unacceptable because it: • cracks finishes; • causes doors/windows to bind; • damages partitions/cladding; • ponds water; • feels bouncy; • looks visibly sagged.

Worked examples

Example 5.1 — Deflection under UDL For a simply supported beam: $\delta_{max} = 5wL^4/(384EI)$

Compare the effect of: • increasing span from 4 m to 5 m; • increasing depth by 25%; • using a higher steel grade without changing section geometry.

Explain why increased steel grade does not materially alter elastic deflection.

Example 5.2 — SLS note Take one beam example and write: • likely ULS concern; • likely SLS concern; • what physical consequence each check addresses.

Resources • Hibbeler Mechanics of Materials: beam deflection chapters. • Jeff Hanson: beam deflection. • SCI, Concrete Centre, and TRADA guidance on deflection limits later.

Gate You can: • calculate standard beam deflections; • explain why span is so influential; • distinguish stiffness from strength; • explain why ULS and SLS are both required.

Module 6 — Failure modes, buckling, restraint and stability

Suggested time: 10–12 weeks

Learn

Material-related failure • yielding; • tensile rupture; • compression crushing; • shear failure; • bearing failure; • fatigue; • brittle fracture: awareness.

Stability-related failure • local buckling; • flexural buckling; • torsional buckling; • flexural-torsional buckling; • lateral torsional buckling; • plate buckling; • global frame sway; • overturning; • sliding.

Buckling fundamentals • Euler buckling; • effective length; • end restraints; • slenderness; • imperfection sensitivity; • why increasing area alone may not efficiently solve a buckling problem.

LTB • compression flange moves laterally; • member twists; • beam loses bending capacity before reaching full section resistance; • restraint must restrain the relevant flange and/or torsional rotation; • construction sequence may change restraint.

Building stability • braced frames; • moment frames; • shear walls; • diaphragms; • ties; • load paths under wind/lateral loads.

Worked examples

Example 6.1 — Euler column comparison Compare pinned-pinned and fixed-fixed columns of equal section and length.

Use: $P_{cr} = \pi^2EI/L_{cr}^2$

Explain what changes when effective length reduces.

Example 6.2 — LTB sketch Take a long steel beam under gravity load.

Draw: • tension flange; • compression flange; • sideways compression-flange movement; • twisting; • possible restraint points from slab/deck/purlins; • torsional restraint at supports.

Example 6.3 — Frame stability Draw a simple single-storey frame:

1. no bracing;

2. diagonal bracing;
3. moment frame.

Apply a horizontal load and explain what prevents sway in each case.

Example 6.4 — Temporary versus permanent stability Identify a temporary condition where a completed structure is stable but an intermediate stage is not, for example: • steel frame before permanent bracing/floor diaphragm is installed; • masonry wall before floor/roof restraint; • slab during wet concrete loading and before strength gain; • precast elements before permanent tying.

Resources • Hibbeler Mechanics of Materials: columns/buckling. • Jeff Hanson: column buckling. • SteelConstruction.info: LTB, bracing, steel stability. • IStructE/SCI temporary stability material where available.

Gate You can: • list credible failure modes for a beam, column, frame, connection, wall, and foundation; • distinguish strength failure from stability failure; • explain LTB in your own words; • identify effective length/restraint assumptions; • identify a complete lateral load path.

Module 7 — Basis of design, actions and Eurocode philosophy

Suggested time: 8–10 weeks

Learn

Design philosophy • characteristic value; • design value; • action effect; • resistance; • partial factors; • limit states; • design situations; • reliability; • National Annexes; • assumptions and model uncertainty.

Actions • permanent actions; • imposed actions; • snow; • wind; • construction loads; • accidental actions; • thermal effects: awareness; • earth/water pressures: awareness.

Limit states • ULS; • SLS; • rare/frequent/quasi-permanent combinations where relevant; • combinations and accompanying actions.

Calculation workflow

1. define structural system;
2. establish actions;
3. identify combinations;
4. calculate design action effects;
5. determine design resistance;
6. compare;
7. check serviceability;
8. provide detailing/restraint requirements;
9. document assumptions and limitations.

Worked example

Example 7.1 — Floor beam from characteristic action to ULS/SLS Use the floor beam from Module 1.

Do the following using the currently applicable standards and National Annex: • classify permanent and imposed actions; • add beam self-weight; • establish basic ULS design line load; • establish basic SLS line load; • calculate ULS moment/shear; • calculate SLS deflection action; • state which checks the line loads will be used for.

Do not worry yet about completing EC3 resistance checks. The purpose is to understand the action side of the design.

Resources • Current EC0, EC1, National Annexes. • SCI/IStructE/Concrete Centre introductory Eurocode material. • SteelConstruction.info searches: • Eurocode load combinations • ULS SLS structural steel design • actions on buildings Eurocode

Gate You can explain: • why characteristic actions are not automatically the same as design actions; • why ULS and SLS use different combinations; • what “design action effect” and “design resistance” mean; • how to set out a simple design basis.

Module 8 — Basic steel design to EC3: laterally restrained beams

Suggested time: 12–16 weeks

This is your Year 1 capstone.

Learn

Section selection • section tables; • steel grades; • self-weight; • section classification; • plastic/elastic resistance concepts; • bending resistance; • shear resistance; • deflection; • bearing/web effects; • laterally restrained assumption; • simple support/connection concepts.

Construction and detailing awareness • bearing length; • web stiffeners: awareness; • padstones; • minimum practical connection information; • corrosion/fire requirements; • fabrication and erection tolerances; • what “laterally restrained” really requires.

Capstone worked project

Brief A laterally restrained simply supported steel beam spans 5.0 m and supports a floor over a 3.0 m tributary width.

For learning purposes: • define reasonable permanent and imposed floor actions from an appropriate standard source; • include self-weight; • assume simple supports; • define a credible compression-flange restraint condition.

Complete:

1. design brief and scope;
2. annotated sketch;
3. load path;
4. characteristic actions;
5. ULS/SLS combinations;
6. reactions;
7. moment/shear;
8. trial UB selection;
9. section classification;
10. bending resistance;
11. shear resistance;
12. deflection;
13. bearing/support considerations;
14. restraint assumption;
15. simple connection concept;
16. conclusion;
17. assumptions, limitations, and required checking.

Resources • SCI Steelwork Design Guide to EC3. • SCI P363 if accessible. • SteelConstruction.info. • An anonymised, checked office beam calculation—ideal if you can obtain one responsibly. • Search: • restrained steel beam design EC3 worked example • site:steelconstruction.info beam design Eurocode 3 • steel beam bearing web buckling guidance

Gate You have completed this module when you can explain every item in your calculation note, including: • why the line loads are correct; • why the beam is assumed restrained; • whether bending, shear, or deflection governs; • what the support/bearing arrangement needs; • what could make your calculation invalid; • what must be independently checked before use.

Module 9 — Deeper steel design: columns, LTB, connections and frames

Suggested time: 16–20 weeks

Learn

Beams • unrestrained beams; • LTB calculation workflow; • restraint spacing; • load-height effects; • destabilising loads; • web bearing/buckling; • stiffeners; • beam-to-column interfaces.

Columns • axial compression; • buckling curves; • slenderness; • effective length; • braced/unbraced frames; • beam-columns and interaction.

Connections • bolted connections; • weld basics; • bearing, shear, tension, block tearing: awareness; • simple versus moment-resisting connections; • connection stiffness; • assumptions in analysis; • base plates: introductory level.

Frames • portal frame basics; • bracing systems; • sway stability; • second-order effects: introductory awareness.

Worked projects

Project 9.1 — Unrestrained beam Take your Module 8 beam and remove the continuous lateral restraint assumption.

Identify: • compression flange; • unrestrained length; • restraint points; • why LTB may govern; • what design/detailing alternatives could improve capacity.

Project 9.2 — Compression member Use a simple steel column under axial compression: • select trial section; • establish effective length; • calculate slenderness; • follow an EC3 column buckling check from a recognised example; • explain why the result is lower than squash-load capacity.

Project 9.3 — Braced steel frame Analyse a simple braced frame: • vertical load path; • lateral load path; • bracing force concept; • column loads; • beam/column idealisations; • construction-stage stability.

Gate You can: • distinguish a beam strength problem from an LTB problem; • distinguish material crushing/yielding from column buckling; • identify whether the global stability system is credible; • understand why connection assumptions affect the whole model.

Module 10 — Timber design to EC5

Suggested time: 12–16 weeks

Learn • solid timber, glulam, LVL, CLT: introductory behaviour; • strength classes; • grain direction; • service classes; • load-duration effects; • bending; • shear; • compression perpendicular to grain; • bearing; • tension perpendicular to grain; • notches and holes; • lateral restraint; • connections; • moisture, decay and durability; • fire: introductory awareness.

Worked projects

Project 10.1 — Timber floor joist Design a simple floor joist: • tributary width; • permanent/imposed loading; • bending; • shear; • deflection; • bearing; • notches and holes; • restraint; • service class/load duration.

Project 10.2 — Timber lintel/beam Assess a timber beam over an opening: • load path; • bearings; • deflection; • wall interaction; • temporary support considerations.

Resources • TRADA. • Wood Campus. • Current EC5 and National Annex. • Search: • Wood Campus timber joist design • TRADA EC5 timber beam worked example • timber notch design shear UK

Gate You can explain: • why timber is not isotropic; • why bearing perpendicular to grain matters; • why notches near supports can be critical; • why moisture/service class/load duration affect design values; • how to check a simple timber member.

Module 11 — Masonry, lintels and existing building alterations

Suggested time: 10–14 weeks

Learn • masonry units, mortar, wall types; • compression-dominated behaviour; • low tensile capacity; • concentrated loads; • bearing and padstones; • lateral restraint; • slenderness; • openings and lintels; • cavity walls; • movement joints; • cracking; • existing structure uncertainty; • surveys; • hidden conditions; • temporary support/propping; • foundation implications.

Worked project

Project 11.1 — New opening in load-bearing masonry Use a textbook/recognised worked example or anonymised office calculation.

Assess: • existing wall arrangement; • loading onto wall; • beam loading; • bearing lengths; • padstone need; • wall compression/bearing considerations; • effects on lower storey walls/foundations; • lateral restraint; • temporary propping concept; • construction sequencing; • unknown-information risk.

Resources • IStructE guidance, where available. • Masonry industry technical guidance. • EC6 and National Annex through work/library. • Search: • masonry padstone bearing structural design UK • existing masonry opening structural assessment • masonry wall lateral restraint guidance

Gate You can explain why “sizing the steel beam” alone is not a complete alteration design.

You should be able to list: • beam; • bearings; • wall capacity; • padstone; • support below; • foundations; • lateral restraint; • temporary works; • survey uncertainty; • construction sequence.

Module 12 — Soil mechanics and geotechnical fundamentals

Suggested time: 18–24 weeks

Learn

Soil types and description • gravel; • sand; • silt; • clay; • organic soils; • made ground; • grading; • plasticity; • density; • water content.

Stress in soil • total stress; • pore-water pressure; • effective stress: $\sigma' = \sigma - u$ • why effective stress controls many soil strength/deformation behaviours.

Water and drainage • permeability; • groundwater; • seepage; • drained versus undrained behaviour; • short-term versus long-term conditions.

Shear strength • friction; • cohesion; • Mohr-Coulomb relationship: $\tau = c' + \sigma' \tan \varphi'$ • what each term represents physically; • limitations of the model.

Settlement • immediate settlement; • consolidation; • differential settlement; • why settlement often governs even where bearing resistance is adequate.

Ground investigation • desk study; • walkover; • boreholes; • trial pits; • CPT/SPT: awareness; • factual versus interpretative reports; • ground model; • uncertainty; • contamination, utilities, trees, drainage, adjacent structures.

Worked examples

Example 12.1 — Effective stress At a given depth, calculate: • total vertical stress; • pore-water pressure; • effective vertical stress;

for a soil profile with groundwater at a stated level.

Example 12.2 — Shear strength concept Use supplied c′, φ′, and effective normal stress values to calculate shear strength from the Mohr-Coulomb equation.

Then explain why the input parameters cannot be guessed for real design.

Example 12.3 — Ground investigation reading exercise Find a factual ground investigation report through a public authority/university search.

Identify: • site location and history; • borehole/trial pit locations; • strata; • groundwater observations; • testing; • made ground; • limitations; • whether it provides design parameters or only factual information.

Resources • Craig's Soil Mechanics. • Das, Principles of Geotechnical Engineering. • ICE and CIRIA resources. • Search: • "factual ground investigation report" pdf • "geotechnical interpretive report" pdf • effective stress worked example soil mechanics • site:gov.uk ground investigation report PDF

Gate You can: • explain effective stress in plain English; • explain why groundwater matters; • distinguish drained/undrained conditions at a basic level; • distinguish bearing failure from settlement; • read a basic GI report critically; • state when a geotechnical engineer must be involved.

Module 13 — Foundations and retaining structures

Suggested time: 16–20 weeks

Learn

Foundations • strip foundations; • pad foundations; • raft foundations; • piles: introductory awareness; • ground beams; • underpinning: awareness; • existing foundations and uncertainty.

Foundation checks • bearing resistance; • settlement; • eccentricity; • contact pressure; • sliding; • overturning; • structural bending; • one-way shear; • punching shear; • reinforcement detailing; • drainage and adjacent excavations.

Retaining structures • lateral earth pressures; • active/passive/at-rest concepts; • drainage; • sliding; • overturning; • bearing; • global stability; • surcharge; • water pressure; • buried services/adjacent structures.

Worked project

Project 13.1 — RC pad foundation Using credible supplied ground parameters from a textbook or checked training example: • establish column load and moment; • choose trial pad dimensions; • calculate bearing pressure; • assess eccentricity; • consider bearing and settlement limitations; • check pad bending; • check one-way shear; • check punching shear; • prepare reinforcement sketch; • identify what geotechnical information would be required in real life.

Resources • Concrete Centre for RC foundation/punching material. • EC7/EC2 resources via employer/library. • Craig/Das for underlying soil concepts. • Search: • pad foundation design Eurocode worked example • reinforced concrete pad foundation punching shear worked example • foundation eccentricity bearing pressure example

Gate You can explain: • why average pressure is not enough; • why a safe foundation must address soil and structural checks; • why settlement may govern; • why foundation design relies on credible ground information; • why retaining walls require drainage and global stability consideration.

Module 14 — Reinforced concrete design to EC2

Suggested time: 20–24 weeks

Learn

RC behaviour • concrete compression; • concrete tension/cracking; • steel reinforcement; • strain compatibility; • neutral axis; • lever arm; • under-reinforced versus over-reinforced behaviour; • ductility.

RC beams • flexural design; • singly reinforced sections; • doubly reinforced sections: awareness; • shear; • links/stirrups; • anchorage; • laps; • curtailment; • cover; • deflection; • cracking.

Slabs and columns • one-way slabs; • two-way action: introduction; • punching shear; • columns and slenderness: introduction.

Durability and construction • exposure classes; • fire; • cover; • compaction; • curing; • propping and striking; • early-age loading; • construction joints.

Worked projects

Project 14.1 — RC beam Design a simply supported RC beam: • actions and combinations; • preliminary sizing; • bending reinforcement; • shear reinforcement; • serviceability; • cover/exposure; • anchorage; • detailing sketch.

Project 14.2 — RC slab strip Design a one-way slab strip: • tributary width; • moment; • reinforcement; • shear; • deflection/cracking awareness; • support detailing.

Project 14.3 — Punching shear awareness Use a flat slab/column example from a recognised text to understand: • critical perimeter; • why punching occurs; • why local thickening or reinforcement may be needed.

Resources • Mosley, Bungey and Hulse. • Concrete Centre. • EC2 and National Annex. • Search: • Concrete Centre EC2 beam worked example • reinforced concrete beam shear links worked example • EC2 punching shear worked example

Gate You can explain: • why bottom reinforcement is used at midspan in a simply supported beam; • why top reinforcement may be needed over supports; • why links are required; • why detailing and anchorage are design-critical; • why concrete cannot be treated as just “steel design with different factors.”

Module 15 — Structural analysis beyond simple beams

Suggested time: 14–18 weeks

Learn • trusses; • plane frames; • portal frames; • arches: introductory; • continuous beams; • moment distribution: awareness; • stiffness method concept; • determinate versus indeterminate structures; • compatibility; • redistribution: awareness; • influence lines: introductory; • second-order effects; • imperfections; • dynamic effects: introductory.

Worked examples • continuous two-span beam; • simple truss reactions/member forces; • simple portal frame under vertical and lateral load; • compare determinate hand result with software result.

Resources • Hibbeler Structural Analysis. • Jeff Hanson, where relevant. • Employer software training and checked examples.

Gate You can: • identify whether an analysis is determinate; • understand why indeterminate structures redistribute force according to stiffness; • explain why connection stiffness/restraint changes force distribution; • perform basic hand validation of a simple model.

Module 16 — Software modelling and model validation

Suggested time: 10–14 weeks

Learn • idealisation; • member local axes; • releases; • supports; • springs; • load cases; • combinations; • offsets; • eccentricities; • diaphragm assumptions; • bracing; • 2D/3D models; • meshing: awareness; • second-order analysis; • model warnings; • reactions checks; • sensitivity checks; • hand checks.

Worked project

Project 16.1 — Beam hand check versus software Model a simple beam in your available software.

Then: • calculate reactions/moment/shear manually; • compare results; • deliberately change a release; • deliberately remove a restraint; • inspect the changed behaviour; • write why the model changed.

Project 16.2 — Braced frame model Model a simple braced frame: • verify vertical reactions; • apply lateral load; • identify bracing force; • check deflected shape; • compare with expected physical behaviour.

Gate You can ask every time: • What is the load path? • What stops translation? • What stops rotation? • Does the model's restraint exist in real life? • Do reactions balance applied load? • Is the deformed shape plausible? • What could the model be missing?

Module 17 — Design detailing, drawings and construction interface

Suggested time: ongoing; initial focus 8–12 weeks

Learn • drawing reading; • member schedules; • GA drawings; • connection details; • reinforcement drawings; • timber details; • tolerances; • bearings; • movement joints; • temporary stability; • erection sequence; • interfaces with architecture/MEP; • RFIs and technical queries; • design changes; • inspection and hold points.

Practical exercises Take anonymised drawings from work, where permitted, and ask: • what is the structural system? • what is the load path? • what assumptions are embedded in this detail? • how is lateral restraint provided? • how is the design built? • what could be misinterpreted on site?

Gate You can identify where a calculation requires a drawing/detail to make it true.

For example: • a beam "restrained by floor" needs a real restraint detail; • a pinned connection needs a connection that does not unintentionally rely on moment continuity; • a foundation calculation needs excavation, bearing, reinforcement, and concrete details; • a timber check needs clear assumptions for notches, holes, connections, and moisture exposure.

Module 18 — Existing structures, surveys and uncertainty

Suggested time: 8–12 weeks

Learn • existing drawings versus reality; • visual surveys; • intrusive investigation; • material identification; • deterioration; • corrosion; • timber rot; • masonry cracking; • movement; • hidden supports; • unknown foundations; • temporary stability during alterations; • conservative assumptions; • when not to assume; • monitoring and specialist survey.

Worked exercise Take an anonymised historic alteration project or published case study.

Prepare: • information required before design; • survey checklist; • uncertainties; • potential load path; • likely risks; • temporary works considerations; • when to involve geotechnical/specialist input.

Gate You can state clearly: "I cannot safely finalise this design until I know X, Y, and Z."

That is good engineering judgement, not a weakness.

Module 19 — Professional practice, checking and business readiness

Suggested time: ongoing

Learn • scope definition; • fee proposals; • exclusions; • appointments/contracts; • professional indemnity awareness; • design responsibility; • checking levels; • calculation checking; • drawing checking; • design risk management; • document control; • CPD; • competence assessment; • client communication; • contractor communication; • quality management; • limitations and reliance on specialist information.

Practical skill For any project, be able to write:

Scope What you are designing.

Exclusions What you are not designing.

Information required What must be provided before design.

Assumptions What you have assumed.

Constraints What site/construction conditions must be met.

Checking Who reviews the work and at what level.

Escalation When specialist or senior review is required.

Gate You are ready for more independent responsibility only when you can: • define a safe scope; • recognise work beyond your competence; • communicate uncertainty; • arrange checking; • avoid accepting unbounded liability for unknown conditions.

Integrated project milestones

These are your proof points. Do not merely complete chapters.

Milestone A — Structural mechanics portfolio Complete and file: • 10 equilibrium/reaction problems; • 10 SFD/BMD problems; • 5 bending-stress/section-property problems; • 5 deflection problems; • 5 buckling/stability sketches.

Pass condition: You can explain each one without notes.

Milestone B — First EC3 beam package Produce a complete learning calculation package for a laterally restrained beam.

Pass condition: A senior engineer can read it and see: • coherent assumptions; • correct workflow; • sensible loading; • credible restraint; • understandable conclusion.

Milestone C — Construction-stage case study Choose a real or historic temporary works situation.

Prepare: • permanent condition; • temporary condition; • load path; • stability system; • sequencing risk; • controls/checks.

Pass condition: You can articulate why the temporary condition may govern even where the permanent structure is safe.

Milestone D — Timber/masonry alteration study Produce a learning case for a straightforward opening or timber floor beam.

Pass condition: You identify the whole system, not just the new beam.

Milestone E — Foundation and ground report review Read one factual ground investigation report and prepare: • ground profile; • groundwater observations; • limitations; • implications for foundation design; • questions requiring a geotechnical engineer.

Milestone F — Integrated small-building scheme Complete a hypothetical small extension/alteration package: • load path from roof/floor to ground; • actions and combinations; • steel/timber/masonry elements; • foundation concept; • temporary support concept; • design risk register; • assumptions and limitations; • list of independent checks/specialist input.

This is a learning portfolio, not construction-ready work.

How to get real problems and checked answers

Source 1 — Textbooks Best for fundamentals because they provide legitimate worked solutions.

Use: • Hibbeler for mechanics/analysis; • Mosley et al. for RC; • Craig/Das for geotechnics; • SCI/TRADA publications for practical design.

Source 2 — University material Search: • site:ac.uk structural analysis tutorial solutions pdf • site:ac.uk Eurocode 3 worked example pdf • site:ac.uk reinforced concrete design worked example pdf • site:ac.uk soil mechanics tutorial answers pdf • site:edu mechanics of materials worked examples

Use material published by known universities and always check the standard/version context.

Source 3 — Professional technical bodies Search: • site:steelconstruction.info EC3 beam • site:concretecentre.com EC2 beam • site:woodcampus.co.uk EC5 • site:istructe.org structural design • site:ciria.org geotechnical • site:ice.org.uk ground investigation

Source 4 — Anonymised office examples This is probably the most valuable source once you have basic theory.

Ask a senior colleague narrowly: “I am rebuilding my permanent-works fundamentals. Could I have one old, anonymised, straightforward beam/foundation/timber calculation to attempt for CPD? I will try it first and then ask for ten minutes of feedback on the assumptions and governing checks.”

Look for: • checked calculations; • accompanying drawings; • site photographs; • design change/RFI history; • construction sequence notes.

That lets you see the full story: not just an equation, but why the engineer made the assumptions.

Source 5 — Your own work For each live project, make private learning sketches only where permitted: • load path; • restraint; • assumed effective lengths; • construction sequence; • failure modes; • what changes at each stage.

Do not remove confidential documents or use client information outside permitted workplace systems.

Your first 12-month sequence

Months 1–2 • Module 0: units, forces, moments, vectors. • Module 1: equilibrium, supports, reactions.

Months 3–4 • Module 1: tributary loading/load paths. • Module 2: SFDs, BMDs, deflected shapes.

Months 5–6 • Module 3: stress, strain, Young’s modulus, material behaviour. • Module 4: centroids, I, Z, major/minor axes.

Months 7–8 • Module 4: plastic modulus, bending theory. • Module 5: deflection and SLS. • Module 6: buckling, effective length, LTB, bracing.

Months 9–10 • Module 7: EC0/EC1 design philosophy, actions, ULS/SLS. • Begin looking at EC3 beam examples.

Months 11–12 • Module 8: laterally restrained EC3 steel beam project. • Produce your first complete calculation package. • Ask for review where possible.

What success looks like after one year

After genuinely completing the first-year programme, you should be able to say:

“I can take a simple structural arrangement and identify the load path, support conditions, internal actions, deflected shape, tension/compression zones, likely failure modes, and stability requirements. I can establish basic ULS/SLS actions and work through a straightforward, laterally restrained Eurocode steel beam design while understanding the purpose and limitations of each check.”

That would be a major improvement in your structural foundation.

It would not yet make you ready for unrestricted independent structural consultancy. But it would make you a far stronger temporary works engineer and give you the correct base for progressing into steel, timber, masonry, foundations, concrete, and eventually carefully scoped permanent works.